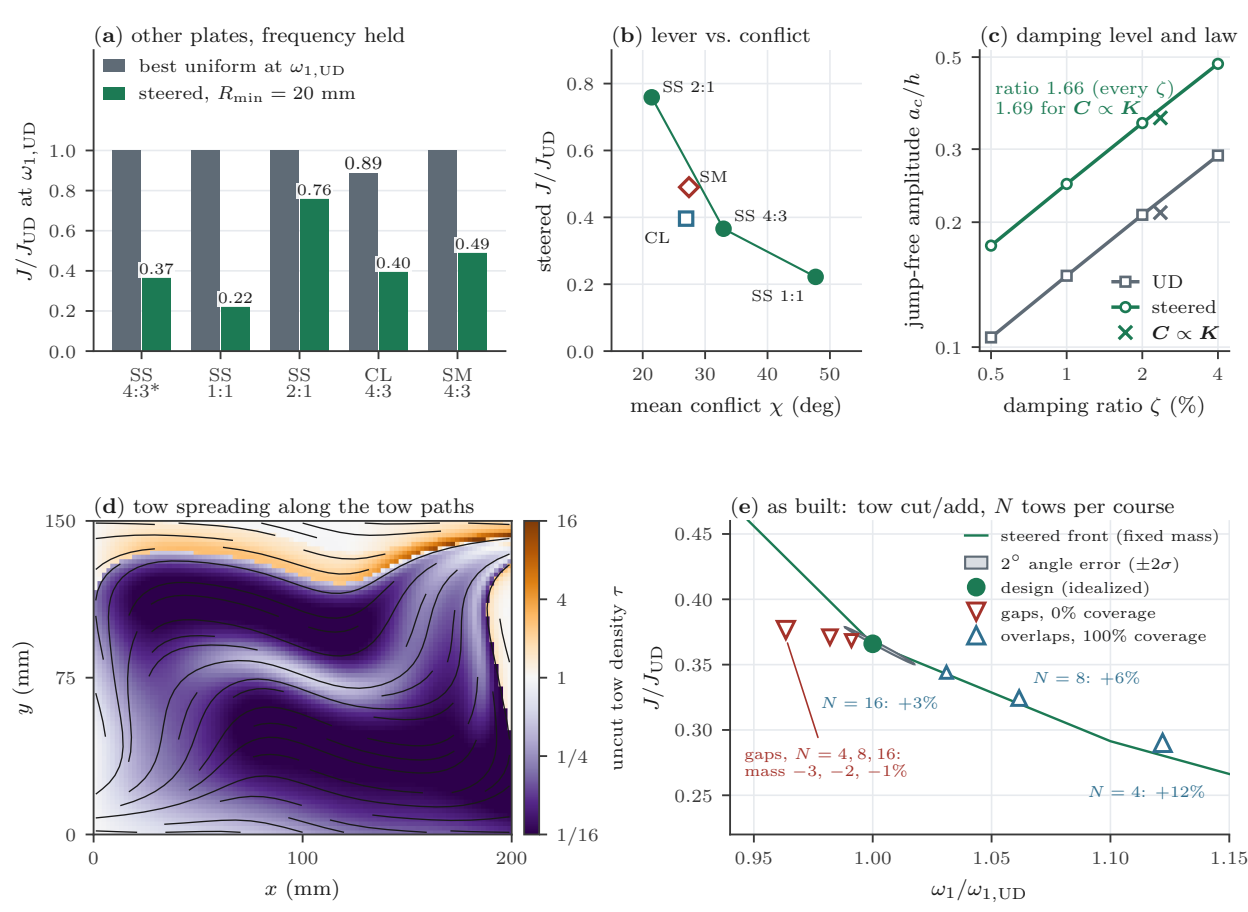

(a) other plates, frequency held
best uniform at $\omega_{1,\mathrm{UD}}$
steered, $R_{\min} = 20$ mm
J/J_{UD} at $\omega_{1,\mathrm{UD}}$
0.37
0.22
0.76
0.89
0.40
0.49
SS 4:3*
SS 1:1
SS 2:1
CL 4:3
SM 4:3
(b) lever vs. conflict
steered J/J_{UD}
SS 2:1
SM
SS 4:3
CL
SS 1:1
mean conflict χ (deg)
(c) damping level and law
jump-free amplitude a_c/h
ratio 1.66 (every ζ)
1.69 for $\boldsymbol{C} \propto \boldsymbol{K}$
UD
steered
$\boldsymbol{C} \propto \boldsymbol{K}$
damping ratio ζ (%)
(d) tow spreading along the tow paths
y (mm)
x (mm)
uncut tow density τ
(e) as built: tow cut/add, N tows per course
steered front (fixed mass)
2° angle error ($\pm 2\sigma$)
design (idealized)
gaps, 0% coverage
overlaps, 100% coverage
$N = 8$: +6%
$N = 16$: +3%
gaps, $N = 4, 8, 16$:
mass −3, −2, −1%
$N = 4$: +12%
J/J_{UD}
$\omega_1/\omega_{1,\mathrm{UD}}$